

In this paper we demonstrate that teams may use genre systems—sequences of interrelated communicative actions—deliberately or habitually, to structure their collaboration. Using data over a seven-month period from three teams' use of a collaborative electronic technology, Team Room, we illustrate that genre systems are a means of structuring six dimensions of communicative interaction: purpose (why), content (what), participants (who/m), form (how), time (when), and place (where). We suggest that researchers and users may benefit from explicitly recognizing the role genre systems can play in collaboration and from examining changes in these six dimensions accompanying changes in electronic technology.

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Genre Systems: Structuring Interaction through Communicative Norms

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In recent years we have seen a proliferation of new ways of organizing work within and across firms, including the use of telecommuters, mobile workers and independent contractors or “eLancers” (Kurland & Bailey, 1999; Malone & Laubacher, 1998); globally distributed (or “virtual”) teams (Lipnack & Stamps, 1997; Maznevski & Chudoba, 2000); and

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virtual organizations (Markus et al., 2000). Many firms and individuals have invested in a great deal of information technology to support these new ways of collaborating across time, space, and organizational boundaries. In such an environment it becomes particularly important to understand the influence of new electronic media on communication. A growing body of literature has examined various aspects of this influence (e.g., Constant, et al., 1996; Fulk, 1993; Hinds & Kiesler, 1995; Walther, 1995; Zack & McKenney, 1995). Additional empirical work (both quantitative and qualitative) has focused more specifically on studying social means of facilitating computer-supported cooperative work (e.g., Bikson & Eveland, 1996; Markus & Connolly, 1990; Olson et al., 1993; Orlikowski, 1992; Suchman, 1996).

In this paper, we propose that genre systems—sequences of interrelated communicative actions (Bazerman, 1994; Orlikowski & Yates, 1994)—are important means of structuring collaborative work both tacitly as habitual mechanisms and explicitly as deliberate devices. We illustrate this claim empirically by drawing on data from a field study of teams using Team Room, a collaborative medium produced by Lotus Development Corporation. Genre systems are generally important ways of organizing the temporal, spatial, and social dimensions of interaction. In addition, we believe that genre systems can also be a particularly powerful means of structuring electronic interactions. With geographically and temporally dispersed groups using new electronic media, key dimensions of collaborative work have all shifted: that is, interaction may happen at different times and/or across different places, allowing different groups of individuals to participate, and encouraging different types of social interaction. Next, this paper elaborates on a set of dimensions of genre systems that help us better understand such collaboration. We conclude by discussing some research implications of our elaborated genre system lens and by suggesting that groups seeking to collaborate electronically may find genre systems especially useful in facilitating their cooperative work.

Genres and Genre Systems

While the concept of genre has a long tradition in rhetorical and literary analysis (Bakhtin, 1986), a number of researchers in cultural, rhetorical, and system design studies have recently begun using it to refer to typified social action (Bazerman, 1988; Berkenkotter & Huckin, 1995; Brown & Duguid, 1991; Miller, 1984; Reder & Schwab, 1988). In previous work, we applied this notion of genres to organizational communications (e.g., memos, meetings, reports, training seminars, resumes, and announcements) and examined them as socially recognized types of communicative actions used by organizational members for particular communicative and collaborative purposes (Orlikowski & Yates, 1994). We identified genres by their socially recognized purpose and by their common characteristics of

form. The purpose of a genre is not an individual's private motive for communicating, but a purpose socially constructed and recognized by the relevant organizational community for typical situations (e.g., proposing a project, meeting to review project status). Form refers to observable aspects of the communication, such as medium (e.g., pen and paper, telephone, or face-to-face), structural features (e.g., text formatting devices such as lists and structured fields), and linguistic features (including level of formality, specialized vocabulary, or graphic devices).

A genre established within a particular community serves as an institutionalized template for social interaction—an organizing structure—that influences the ongoing communicative action of members through their use of it within and across their community. Genres as organizing structures shape, but do not determine, how community members engage in everyday social interaction. In many instances individuals draw on existing genre norms out of habit to facilitate a particular communicative act (e.g., tacitly using a standard memo format for interdepartmental communication or a standard agenda for a routine project update meeting). In other instances individuals may draw on genre norms more deliberately to accomplish a communicative action (e.g., intentionally choosing a letter template rather than an informal note genre in composing an electronic mail (e-mail) message addressed to an unfamiliar international correspondent, or explicitly shifting a meeting from face-to-face to audio conferencing to accommodate remote participants). Whether used implicitly or explicitly, genres are a powerful source of communicative norms for social activity (Yates et al., 1999).

In some cases genres are linked or networked together in a way that constitutes a more coordinated communicative process. For example, committee meetings often include a series of genres such as oral presentations, dialogue, and voting. On the other hand, major project contracts are often realized through an interlinked sequence of genres including a request for proposal (RFP) from the initiating organization, a proposal submitted to the initiating organization by a responding organization, an evaluation report prepared within the initiating organization to evaluate the proposal, and a contract between the two organizations (or a rejection letter from the initiating organization to the responding organization). Such a genre system consists of interdependent genres that are enacted in some typical sequence (or limited set of acceptable sequences) in relation to each other, and whose purpose and form typically interlock (Bazerman, 1994). For example, the job ad, job letter and resume, invitation to interview (or rejection letter), interview, and job offer (or rejection letter) form a carefully orchestrated genre system constituting a hiring process. Such systems are composed of a coordinated, interconnected set of communicative actions that together accomplish an interaction (such as hiring personnel). From an organizational standpoint, collaborative activity is

often organized and defined by such genre systems. By examining these genre systems in practice we can learn much about collaboration in general, and distributed, computer-supported collaboration, in particular.

This notion of a genre system as a series of genres comprising a social activity and enacted by all the parties involved is especially useful for studying interaction because it focuses on how people use sequences of communicative actions to coordinate their activity over time and space. We suggest that, like individual genres, genre systems are also organizing structures within a community that provide expectations about the purpose, content, participants, form, time, and place of communicative interaction. In other words, both genres and genre systems carry expectations about why, what, who/m, how, when, and where.¹ For analytic convenience we will treat these dimensions of communicative interaction separately, although in practice they are closely interconnected.

Why

Most importantly, the genre system and its constituent genres provides expectations about its socially recognized purpose. For example, the genre system around awarding project contracts is intended to select the best bid for the firm issuing the RFP. As the genre system becomes contextualized for projects of a particular type (e.g., construction projects) it may have one or more added purposes such as assessing the new construction's impact on local traffic patterns. The proposal itself has at least two purposes: to get the proposed project accepted, and to get the bidding firm hired to carry out the project (Houp & Pearsall, 1977). As with genres, we see the socially recognized purpose, or why, as the central identifying characteristic of genre systems.

What

The genre system also provides expectations about the content of the whole genre system as well as the content of its constituent genres. The genre system around awarding a project provides expectations about which genres typically appear (e.g., the RFP and the contract) and in what possible sequence(s). For example, the RFP is first, and before proposals are actually submitted, an optional bidder's conference may be conducted for discussion and clarification of the project for the potential bidders. These individual genres also carry expectations about their content. For example, the RFP is expected to contain specifications on which proposals should be based, and the proposal is expected to provide a cost estimate and time schedule.

Who

A genre system provides expectations about the participants involved in a communicative interaction and their roles. It specifies who initiates which genres and to whom such genres are addressed. For example, the

firm soliciting bids in a large project initiates the RFP making it available to a more or less limited set of potential bidders, while a responding firm sends its proposal to the firm that solicited bids.

How

A genre system provides expectations about the form of the genre system, including expectations about media, structuring devices, and linguistic elements. For example, the genre system around awarding a small project typically embeds the proposal in a letter format (which could be sent on paper or via an e-mail attachment) and uses less formal language. In contrast, the genre system around awarding a large project creates the expectation of an extended proposal by using formal language, printing it on paper, and binding it into one or more volumes.

When

Participants in a genre system often attach specific temporal expectations, typically stated as deadlines, to different constituent genres. For example, the RFP typically contains a deadline by which proposals must be received to be considered. Moreover, the proposal in this genre system must include a proposed timeline for performing the project. Sometimes the temporal expectations are not explicitly stated but are known to experienced participants.

Where

Finally, a genre system provides location and place expectations for the entire genre system and its specific genres. For example, the proposals must be sent to a specific address, whether physical or in cyberspace, and the bidders' conference takes place in a stated location (whether electronic or not).

To summarize, a genre system, when enacted by participants, structures or choreographs multi-party interactions within and across communities by specifying the key dimensions of communication—purpose, content, participants, form, time, and location. A genre system therefore, serves as an interaction template that participants draw on in engaging with each other across media, time, and space. Of course, by specifying dimensions of communicative interaction, the genre system template also specifies those aspects of interaction that are not sanctioned by the community using the genre system. The expectations about purpose, content, participants, form, time, and location embedded in genre systems thus reveal, for example, who is not empowered to initiate or receive certain genres, when or where certain genres may not be enacted, and what content or form is inappropriate for particular genres. As with all structuring templates, genre systems serve to both enable and constrain types of interaction and modes of involvement. Genre systems are indicative of

what a community does and does not do (purpose), what it does and does not value (content), what different roles members of the community may or may not play (participants), and the conditions (time, place, form) under which interactions should and should not occur.

Genre Systems in Team Room

In the next section we will discuss data from the early use of Team Room by three teams in a high-technology company in the northeastern US (Mox Corporation, a pseudonym) to illustrate how genre systems structure ongoing interaction. In this section, we briefly describe the setting and data we draw on and the analytic methods used (see Yates et al., 1997 for more details), as well as identify the three genre systems that serve as illustrations for our theorizing.

Data and Analysis

The Team Room technology is a collaborative application built within Lotus Notes and designed specifically to support teams in organizational settings (Cole & Johnson, 1995). The Team Room database is organized around two key fields specified by the user for each posting: communication type, which indicates the general purpose of a message (e.g., discussion, reference); and communication category, which indicates the topic (e.g., planning, education). A few default communication types and categories are provided by the software to help teams get started. Developers of the system also created facilities for team members to create their own types and categories when initiating a new database so they could tailor them to suit their context and projects. When a new Team Room is initiated someone on the team must complete a Mission Page specifying types and categories and endorsing defaults before messages can be entered into Team Room. The tool was designed to encourage teams to label the *why* and *what* of their messages, but the teams we studied used this feature inconsistently.

The three teams in our study, which were serving as beta test sites for the Team Room software, were the Information Systems Quality Improvement (ISQI) team, the Information Systems Leadership (IS Leadership) team, and the Philanthropy team, all of Mox Corporation. The first two of these teams were part of the Information Systems (IS) Department and consisted of technically adept members familiar with various forms of electronic communication, including Lotus Notes. The third team consisted of corporate staff with limited technological experience. In total, the three teams posted 492 Team Room messages in the seven-month study period.

The ISQI team, which was composed of three core members plus a larger set of interested parties, used its Team Room the most extensively, posting a total of 238 messages. This team saw its purpose as being to "oversee and drive the implementation of TQM [Total Quality Manage-

ment] in Mox I.S." The team leader initiated the use of Team Room at the team's formation, stating that he saw the groupware technology as "a way of creating a running documentation for the quality improvement project that would also support interdependent work and inclusion of those outside the immediate work circle."

The IS Leadership team included twelve members, each of whom had his or her own subordinates and responsibilities, and several of whom worked remotely at least part of the time. The team's Mission Page defined its role as, "To improve the productivity of Mox by connecting the desktop to corporate data, seeking opportunities to reengineer business processes using technology, and enhancing Mox's ability to perform its business." The team leader believed Team Room would facilitate communication among the members by creating one main channel for communication and simultaneously produce an archive for several long-term projects. This team's use of Team Room generated 188 messages over the period studied.

Finally, the Philanthropy team, composed of five members spanning three levels of hierarchy, described its mission as: "To facilitate the sharing of a portion of the company's profits, products, and people in ways that assist individuals and communities, particularly those racially and economically disadvantaged, in achieving their highest potential in terms of social and economic development." Although the team existed before Team Room was made available, its members tended to work independently rather than collaboratively. The team leader supported the use of Team Room specifically in order to promote collaboration among the members. Plagued by early technical difficulties, plus the absence of compelling collaborative activity, the Philanthropy team generated only 66 postings in the period studied.

Our data for the study consisted of 492 messages posted in the three Team Room databases over a seven-month period as well as interviews with members of the teams. To analyze the Team Room messages we developed a genre coding scheme based on socially recognized purpose (as interpreted from the message content and the designated communication type) and form (including formatting features such as subheadings, and linguistic practices such as salutations and informal language).² In this study we identified genres primarily by purpose, designating relatively few form categories as mandatory for a genre. We believe that most genres of organizational communication are associated with a constellation of form features, some combination of which are necessary for members of a particular community to recognize a communication as an instantiation of that genre. In some cases we specified that certain genres should have at least one or more of a set of form features. In addition to coding and analyzing the messages, we conducted interviews with members of each team at various points during the study. Early interviews

focused on the adoption and initial use of Team Room. Subsequently, after analyzing the messages using the genre coding scheme, we conducted further interviews with key members from each of the teams using a variation of the discourse-based interview (Odell, Goswami & Herrington, 1983). These interviews helped us to understand some of the organizational conditions influencing use of Team Room, allowed us to interpret the genre systems we had identified in it, and informed us about how team members communicated in media other than Team Room (e.g., phone, e-mail).

Three Genre Systems

Based on our analysis of the data, we identified three genre systems used by all three teams: the meeting genre system, the collaborative authoring genre system, and the collaborative repository genre system. Almost half of the messages could be categorized into one of these genre systems.³ We consider this percentage quite high, as we would not expect all or even most such group communication to take the form of these highly choreographed interactions. Rather, we would expect a considerable amount of communication in the medium to instantiate individual genres or even to be idiosyncratic communications.

The meeting genre system involved the communicative activity around face-to-face meetings including meeting logistics, meeting agenda, the meeting itself, and meeting minutes distributed subsequently. The meeting logistics and meeting agenda genres often overlapped, with a single posting containing both. For instance:

REMINDER: TQM MEETING

Thurs Sept 26 from 10 to 11 in Conference Room C

The agenda for Thursday's meeting is:

1. ...
2. ...

Agendas were presented at the beginning of meetings when not distributed in advance. Meetings involved face-to-face interaction. Finally, minutes were kept for some meetings. This meeting genre system is clearly rooted in norms established originally in the paper-based world and subsequently transferred into electronic media.

Next, the collaborative authoring genre system, which centered around the cooperative act of authoring texts included three genres: a circulated draft, responses to this draft, and a final version. The circulated draft was often accompanied by an explicit request for comments. The following example demonstrates this cooperative act around the design of a TQM training course by the ISQI team: "Here is a first shot at a course outline.

... Please comment on what might be missing or over emphasized." Alternatively, the invitation to comment was implied rather than stated, as demonstrated in this subject line to a circulated draft: "[Subject:] Here's the start of communication strategy (work in progress)." The responses to the draft came in two variations—which we refer to as feedback on and dialogue around the draft—though the distinction is not always a precise one. In the feedback-on-draft variant the response focused on the text of the draft itself, making editorial suggestions. For example, in a circulated draft of a customer questionnaire, one team member provided specific feedback on the wording of this question as follows:

2. How well do our applications ~~on the development process~~ meet your needs? Have you ever been involved with Applications Group in the development of an application? (If so,) how well did that process work for you?

In the dialogue-around-draft variant the discussion moved beyond the wording to the subject of the draft, including discussions of policy and implementation. For example, in response to a circulated draft proposing weekly TQM lunch sessions as a means of disseminating ideas for improving quality, one member of the ISQI team offered the following comment about how to implement the suggestion:

We could start initially by having the conversation be open, and then as we get going, bring up different topics. I also think that at first we should all plan to attend (me, Todd, you) and eventually we might be able to rotate chairmanship.

The two variants sometimes became intertwined. In one case, for example, a participant attached a revised version of a circulated draft to a message stating that "My changes reflect a sense of personal accountability rather than staff accountability." The third genre in the system—the final version—was rarely posted in the Team Rooms. We discovered during interviews that final versions were more often produced in another medium such as presentation graphics. The collaborative authoring genre system resembles those conducted in paper media or via e-mail, yet broadens the involvement and interaction among team members, allowing for team (rather than one-to-one) dialogue around issues.

Finally, the collaborative repository genre system involved using one document as a placeholder, which invited subsequent contributions on the stated topic in the form of responses (or "comments," as they are designated in Notes and in Team Room) nested within the placeholder document. The teams we studied used this genre system to support diverse activities such as coordinating schedules, brainstorming, initiating discussions, and consolidating topical information. There were too few examples and too many activities to identify variants of the genre system. Gen-

erally, they ranged from listing vacation schedules in a common place, to brainstorming new ideas. Consider the following example of a placeholder posted in the ISQI Team Room:

[Subject:] Place to Brainstorm on Communication Ideas

Since communication is such a large part of what needs to happen in order for the whole of I.S. to be involved in this quality transformation, I'm opening this document as a placeholder for brainstorms. Compose comments whenever you get inspired.

The responses to this placeholder included simple suggestions such as starting a library of books and articles on quality, as well as more elaborate suggestions that built on a previous suggestion:

[Subject:] More ideas for subversive posters (Quality IS, Quality IS NOT)

I can imagine a whole series of posters on this theme of the subtle and not so subtle differences between what we do now and what we might be doing. They need pictures and need to be short and sweet.

[followed by table with five examples of Quality IS, Quality IS NOT pairings.]

Another response outlined an exercise that the team could run at an upcoming quality lunch. In our data set, responses to a placeholder tended to be parallel (e.g., vacation schedules, some of the communication ideas) or cumulative (e.g., the ideas for "subversive posters," which built on each other) rather than contentious. This genre system makes use of the document-response architecture provided by Team Room (in turn based on that of Lotus Notes) in an innovative way to support activities that in the e-mail or paper world would have happened less systematically or less visibly to all members of the team, if at all.

In some instances these genre systems overlapped. For example, sometimes the person circulating a draft for comment did so by way of a placeholder message asking for comments to be added in responses to the document. Similarly, the agenda of an important meeting could be circulated as a draft for comments. Although the three genre systems are distinct, in practice they sometimes overlap and otherwise blend together.

Genre Systems as Structuring Interaction in Team Room

In this section, we will draw examples from the three genre systems used in the Mox Team Rooms to show how they embody expectations that structure several dimensions of communicative interaction: why, what, who/m, how, when, and where. In some cases we will see explicit awareness of the expectations and their coordinating role, while in others the teams appeared to enact the coordinating genre systems more or less habitually.

Why

Most centrally, a genre system provides expectations about the socially recognized purpose of the system as a whole, and of the genres that compose it. For example, the meeting genre system had as its central purpose the coordination of a meeting among a specified set of people to achieve some purpose. In some cases in the Team Room data, a team member called a meeting for a specific purpose (e.g., meetings by the ISQI team to organize specific TQM events or projects) and used the genre system deliberately to focus attention and coordinate action around that specific purpose. In such cases, for example, the meeting organizer circulated a draft of the logistics and agenda to get input that would make the purpose of the meeting as explicit and attractive as possible to potential attendees. In other cases the meetings were simply regularly scheduled staff meetings (e.g., the ISQI team's weekly or bi-weekly meetings) and team members invoked the genre system unreflectively to coordinate attendance. Within the genre system, the meeting logistics message coordinated the appearance of the desired set of individuals at the meeting, and the meeting minutes, when present, served to document the decisions and commitments team members made at the meeting.

As another illustration, the socially recognized purpose of the collaborative authoring genre system was to solicit suggestions and incorporate them into a document considered important to the team and needing buy-in from other team members. The circulate-draft genre presented the draft to others and requested group input. The response-to-draft genre provided the requested input, either to the document itself (in the feedback-on-document variant) or to the philosophy and policy evident in the document (in the dialogue-around-document variant). The final version, when present, reflected how group members' suggestions had been incorporated into it.

As a final example, the collaborative repository genre system had the general purpose of designating a specific (virtual) place for electronic input about a specific issue. It allowed brainstorming and coordination without a face-to-face meeting. For example, the IS Leadership team, consisting of members with busy schedules and separate areas of responsibility, frequently used this genre system to get the input needed from its members to make scheduling, policy, and strategy decisions. Initially, they probably used this innovative genre system deliberately to help coordinate their input without a meeting, and over time, such use may have become habitual.

What

A genre system provides expectations about the content of the system (i.e., its constituent genres) and of the genres within it. For example, the collaborative authoring genre system was typically composed of the following sequence of genres: a circulated draft, some responses (feedback

on, and dialogue around draft), and, in its most complete form, a final version. The final version of a document appeared in the Team Rooms in only four cases, but interview data revealed that the absence of final versions was not viewed negatively suggesting that the genre system did not carry a strong expectation of such a posting—perhaps because the final version was often made available in some other medium or location.

A particularly interesting issue related to the content of this genre system was what we came to call “orphaned drafts.” Of the 41 circulated drafts posted in the three Team Room databases, 22 received no responses, that is, no team members took up the request for input.⁴ Although these “orphaned drafts” could be interpreted simply as free-standing genres (drafts posted for the information but not action of others), the requests for input made clear that they were intended as elements in a genre system. Interviews shed some light on how team members viewed these drafts. For example, one member of the ISQI team explained that “some of these incomplete documents stem from projects that were put on the back burner.” This member expected such drafts to be taken up later. In contrast, another member of this team had a somewhat different interpretation of such incomplete genre systems:

I think we're an organization of people who basically have attention deficit disorder. If it's not on everyone's frontal lobes at the time, then it's gone. There's no back burner, there's no, “Hmm, that's interesting, we're going to come back to that at some point.” If it's not on fire, it doesn't exist.

Yet another interpretation came from a member of the Philanthropy team who suggested that team members sometimes posted drafts that had already been discussed face-to-face and the absence of any response was taken as agreement that the write-up “represents what had been decided.”

A similar issue of incomplete genre systems arises with regard to the content of the collaborative repository genre system. This genre system typically consisted of a placeholder document and responses to it. In 5 out of 21 cases, however, no responses appeared to a posted placeholder. Again, interpretations of this phenomenon varied. One member of the IS Leadership team, in describing habitual participation in this genre system, viewed responding to placeholders as optional:

I think there's an unstated norm in these placeholders or brainstorming that not everyone needs to comment on it. You only comment on something if you have an opinion. Whereas if there's an action for you, there's a norm that you need to respond to it. But if it's a placeholder . . . if you see something in there and you agree or you don't care, people don't comment on stuff like that.

Despite this belief in optional responses, however, the same member revealed that such behavior had personal consequences:

The way that I feel if I put something in and no one comments on it, it's exactly like saying something in a meeting and having no response around the table. You just feel like—unless you really want to push it—say it again, say it louder, say it to somebody in particular—you just sort of say, “Well, never mind.” I would always get a little perturbed by that. I don't know how other people felt. Anytime I put a document in and no one comments on it, I think it's really rude I guess. . . . It's like saying something out loud and having it go off into the ether. It's a disempowering thing, to be ignored. It's being ignored, basically.

This member's expectations around what genres make up this genre system (a placeholder and some responses) contributed to his negative interpretation of the lack of responses to posted placeholders. That is, he interpreted as deliberate the behavior he had earlier described as habitual.

In contrast, the absence of minutes from an example of the meeting genre was not enough to make participants perceive the genre system as incomplete. As one participant noted in an interview:

Minutes are not a routine thing. Not for staff meetings at all. . . . We *never* do minutes of a staff meeting. More often than not I think people operate under the assumption that if you were there [at the meeting], you took your own notes because you're the only one who knows what you need to take away from that meeting. . . . If you were there, you got it, or you should have gotten it. Having minutes is like having a crutch.

Thus minutes were not seen as a necessary part of the genre system—indeed, use of minutes was deliberate rather than habitual. The absence of minutes would have been notable only if someone had promised to produce them, a situation that did not arise in our data.

Within the genres constituting the genre system, the “what,” or content refers to the subject areas typically covered in the genres. For example, meeting logistics messages were expected to contain information about the time and place of the meeting; agenda messages announced topics to be covered in the meeting; and minutes, when present, summarized decisions and commitments. Expectations around the content of the constituent genres seem to have been clearer than those around the genre systems discussed above.

Who/m

A genre system may carry expectations about roles in communicative interactions that help coordinate collaborative action. In a given genre system the relevant community may identify who can initiate which genres and for whom such genres are intended. In the Mox teams' use of the genre systems the roles are not as clear-cut as in some other familiar genre systems, such as the genre system around awarding contracts, referred to above. Interviews with ISQI team members suggested that the three team

members, though representing two hierarchical levels, were not very hierarchical in their work styles. Thus, different individuals might draft and post documents on different subject areas. Nevertheless, the official team leader of ISQI saw his role as setting direction and encouraging the collaboration of the others in developing that direction. Thus, documents that set direction were typically drafted by him. In the Philanthropy Team, interviews revealed more issues around ownership of projects and initiation of drafts in Team Room, although these were linked more to interpersonal friction than to hierarchy. One person explained:

There was also a thing about establishing a presence with the project. It seems silly, but the person who creates the document, if there is conflict in the group, that becomes something people pay attention to. I started paying attention to who created a document and what that meant. Does that mean that that person is then in charge of the project, and those kinds of things? At the end, does it look like that person is all over the Team Room, like that person is doing more?

The issue of who initiated certain genres, and thus, certain genre systems, had more significance in this group where interpersonal tensions were more evident and where the group worked less as a team than did the ISQI team. Indeed, the quote suggests that expectations around the meaning of initiating certain genre systems set the stage for the playing out of within-group competition for attention and resources.

The genre systems we observed in the use of Team Room carried clearer expectations around "for whom" they were intended than around "who" initiated them. The Team Room medium supported a change from tasks handled in media such as e-mail by making most team messages available to all team members (of course, an individual could still send a personal message by e-mail). This allowed more inclusive coordination. For example, one Philanthropy team member stated in an interview, "I think it's good that everyone has access to [Team Room]. I think it can bring the level of discussion to a different, deeper [level]." The same person noted in another interview that in Team Room she could post a draft to everyone at the same time rather than first clearing it with her boss:

[Team Room] made it possible for me to post something to the group. Without it, I would probably have gone to [the team leader] with the idea, and then discussed it with him, seen what he had to say, then presented it at a staff meeting. But it would have gone through him first. This way sort of bypassed him.

Although this team member could still have presented an idea first to the team leader, she appears to have deliberately chosen to bypass him by posting the draft on Team Room for all to see.

In general, these genres and genre systems as they were used in Team Room, whether deliberately or habitually, increased participation of the whole team and provided a mechanism for coordinating that participation. In cases where the team was less strong and there were tensions and rivalries (Philanthropy), initiation of a genre system involved more explicit strategy (e.g., taking ownership of a project or idea by drafting a document on it) or an explicit bypassing of hierarchical channels. Where the team was more cooperative (ISQI), multiple participants could post the initiating document of a genre system but the team leader always took this role in certain topic areas (reflecting interacting “who/m” and “what” expectations).

How

A genre system also provides expectations about the typical form (including media) of the genre system and its constituent genres. During a given time period, a genre system may be associated with norms about the media acceptable for various constituent genres, though such norms clearly change over time and may be in flux at a particular period in time. For these primarily collocated groups, for example, the meeting genre within the meeting genre system was expected to be face-to-face. For geographically dispersed groups, however, we can imagine such “meetings” taking place via audio or video conferencing or even by synchronous or asynchronous text-based conferencing. The shift from face-to-face communication to another medium would be a significant one that would probably involve some shifts in the other genre norms of the meeting.

The newly introduced Team Room medium was a salient aspect of the “how” expectations of the Mox teams.⁵ In the collaborative authoring genre system, the introduction of the Team Room medium had led to a shift from circulating a draft to selected people (on paper or as an e-mail attached file) to posting it in the Team Room where all other team members could see it. Significantly, this also meant that every member of the team saw every other member’s comments rather than having only the primary author see all of the comments. The interviews indicated that this shift in medium supported more involvement by the team as a whole (an interaction between “how” and “who/m” expectations).

The collaborative repository genre system was based on the structure underlying Team Room and Lotus Notes—the document followed by responses. While some instances of it probably replaced one or more face-to-face meetings (e.g., updates on the outsourcing project), others replaced an e-mail or paper query that would have been responded to by each person individually (e.g., on vacation schedules), and others might not have occurred at all (e.g., brainstorming learnings). As realized in Team Room, the genre system supported easy asynchronous contributions and increased involvement of the whole team. Thus, we can see that genre systems could change when the medium in which they were realized changed. The use of

the medium might even encourage the emergence of new genre systems, as the underlying document-response structure of Notes and Team Room seems to have done for the collaborative repository genre system.

The "how" expectations around individual genres are not as constitutive as are the "why," "what," and "who/m" expectations. They are also not as critical to coordination, but are significant socially recognized norms that can be adopted habitually or deliberately as a way of facilitating coordination. For example, in the three Team Rooms, 13 out of 17 meeting agenda messages had lists as a form feature, and 12 out of 17 meeting minutes had subheadings. These form features could be used by authors of agendas and minutes as easy, habitual forms for presenting the content and/or as ways of deliberately drawing attention to important points. For example, minutes for one meeting of the Philanthropy team contained a section dedicated to "Next Steps" that identified the tasks assigned to various individuals by using their names as subheadings, and thus, highlighting what each person had agreed to do. The placeholder genre in the collaborative repository genre system always took the form of a document, while the response genre always took the form of a comment to a placeholder document. In the collaborative authoring genre system, 20 out of the 41 circulated drafts were drafted in another application (e.g., word processing or spreadsheet), then imported into, linked with, or attached to a message within the Team Room. In this instance, the tool supported such form characteristics as the attached, linked, or embedded file, allowing drafts created in other media to be put into Team Room where responses could be seen by all, thus facilitating coordination.

The expectations for the form of the genre system and its constituent genres were not as strong as those for purpose, content, and participants, but the form expectations were part of the constellation of expectations associated with the system and the genres making it up and, in many cases, served to structure the communicative interaction.

When

Genre systems also structure temporal aspects of coordination. Initiators of genre systems or of the genres constituting them may tacitly accept timing assumptions already associated with the genre system, or may deliberately shape them as coordination devices. For example, the ISQI team often used the meeting genre system around its regular team meetings. The genre system created expectations for sequence and timing of the constituent genres. The meeting organizer's announcement of the day, place, time, and general purpose of the meeting typically initiated the genre system. Initially, the meetings occurred every week, but later, as the project task shifted, the team deliberately shifted the meeting schedules from weekly to bi-weekly, and over time, even less frequently. The genre systems could also be used to shape the opportune time (Yates &

Orlikowski, forthcoming) for some communicative action. We have noted that minutes were not typically used for such regular staff meetings. Nevertheless, minutes for some of the ISQI group's regular meetings as well as special meetings were posted in the Team Room. A team member explained in an interview when minutes might be posted: "Either it had been a very productive meeting and we thought that we needed to capture what we had gleaned from the meeting, or . . . we needed to share some agreements we'd made with other people." Thus minutes, the last and optional genre of the complete genre system, resulted when a member deliberately recognized and/or shaped an appropriate moment to produce minutes.

Although the uses of timing just cited were deliberate, we see cases in which the team members accepted certain expectations about timing quite tacitly. In the collaborative authoring genre system we have noted that some drafts were orphaned—that is, no one commented on them to complete the genre system. In most cases when any reactions to a particular draft were posted, the first reaction came within about a week after the draft's posting. This week appears to reflect a tacit "window of opportunity" (Tyre & Orlikowski, 1994) for responses. After that period responses were less likely to appear, as the appropriate moment had passed. We saw evidence, however, that the team member who posted the draft also had the ability to shape the opportune moment (Yates & Orlikowski, forthcoming). One team member commented on the orphaned drafts as follows:

It's how things work at Mox. . . . [I]n this kind of environment . . . people just sort of float an idea out there and see if it takes off. If no one responds to it, then you have to decide, 'Is this something I care about enough to push through the organization or would I rather put my energy elsewhere?'

Although this explanation accounts for the fact that many of the orphaned drafts were simply dropped, it also allows the initiator who "care[s] enough [about something in the draft] to push [it forward] through the organization." In doing so, he or she would extend the window of opportunity, rather than accepting the time expectations carried by the genre system.

The shift in medium (how) can also create the opportunity for shifts in "when." For example, by posting drafts in Team Room, authors made them instantly available to the whole group. While an individual might have procrastinated in commenting on a draft sent by mail or e-mail, the appearance of comments on the draft in the group's Team Room encouraged prompt responses as part of the dialogue around draft.

Where

Finally, a genre system structures location and place expectations for the entire genre system as well as for specific constituent genres.

Although genre systems and constituent genres carry generalized cultural expectations about place, each instance is inherently bound by unique situational conditions. For example, a member of the ISQI group explained that another member, initially reluctant to use Team Room rather than e-mail for much of the team's communication, had changed his view by the end of the most active period of use, "he got why we were doing this because it was so valuable to go back in and have one place where that stuff was held."

Similarly, another member of ISQI noted that,

We've used Team Room as the place to make, put words on all the individual thoughts we've been having. And to put them out there for comment from other people. So, it's been a place where I'll start forming an idea, and put it out in whatever ill-formed shape I have as a discussion document, and I'll request as part of that, that [my colleagues] talk about it.

So when individual team members enacted innovative genre systems such as the collaborative repository genre system in Team Room rather than in e-mail, the team as a whole gained a clearer sense of place for its activities. This structuring of team place was perceived as an advantage by users allowing them to coordinate their activities through collocation in an electronic space.

Although all three of the teams were colocated, the advantage of using an electronic space to constitute a team place was evident when members not physically present were added. A member of the ISQI team noted that usage of Team Room

has picked up lately because we have a new quality office member in Europe and she is—that's pretty much the only way we can communicate with her except for via e-mail—and she's putting a lot of stuff in there. So that's sort of encouraging us to do that.

But the tool by itself was not enough to create a sense of place—team members had to use it in ways that structured their electronic space to produce what might be called a "team place," as the members did when they used genre systems such as the collaborative repository. The same ISQI member talked of another Team Room with distributed members that was never used successfully because people simply did not respond to drafts or placeholders posted in it—that is, they did not use the electronic space to construct a shared place for their team activities. Similarly, the Philanthropy group, though it had no dispersed members, seems never to have created a space for genuine collaboration, with members instead focusing primarily on political aspects of use.

The placeholder genre most explicitly attempted to create a space for communal thinking. It built on the spatial quality of Team Room, inviting

others to join in populating the space within it. Even the simplest use of this genre system, for posting vacation schedules ("Please enter Vacation Plans for July and August") invited others to enter the space it was attempting to create.

Conclusions and Implications

As this paper has shown, genre systems play an important role in structuring expectations around several dimensions of communication: *why*, *what*, *who/m*, *how*, *when*, and *where*. That is, they structure expectations about the purpose, content, participants, form, time, and location of communicative interaction among members of a community.

Our study at Mox corporation indicates that the teams enacted at least three genre systems—meeting, collaborative authoring, and collaborative repository—within the new Team Room medium. In examining how the various dimensions of the genre systems structured the teams' interaction over time, we were also interested to see how the use of the new medium had facilitated changes in interaction. Our analysis suggests that the enactment of the three genre systems within Team Room, both reinforced and changed—whether subtly through enhancements or, more significantly, through transformations—the teams' communicative practices. Table 1 depicts the six dimensions of communicative interaction in the three genre systems enacted within Team Room, and whether the associated expectations were the same as, or different from, those around the genre systems as they would have been enacted on paper or in e-mail. We have also indicated the consequences for the teams' existing communicative practices (i.e., reinforcing, enhancing, or transforming).

In particular, we find that the meeting genre system, as enacted within Team Room, essentially preserves and reinforces the existing communicative practices around meeting logistics, agendas, and minutes, as enacted on paper or in e-mail. The collaborative authoring genre system reinforced the existing communicative practices around drafts, responses, and final versions, while also enhancing dimensions of this communication through facilitating changes in when, where, and how the interaction for collaborative authoring occurs, as well as potentially allowing different types of content (e.g., dialogue around draft) and participants to be considered and involved. Finally, the collaborative repository genre system, with its placeholder and response genres, carries different expectations about some dimensions of the teams' communication (how, when, and where) and potentially different expectations about others (why, what, who/m), which represents a departure from existing communicative practices of the teams. As such, it potentially transforms the interaction of the team members around such activities as coordination, scheduling, brainstorming, decision-making, and synthesis. We have noted that these changes are potentially transformative because the pur-

Table 1
Reinforcement and Change of Mox Communicative Practices in the
Enactment of Genre Systems Associated with Use of Team Room
Rather than Paper or E-mail

	Meeting	Collaborative Authoring	Collaborative Repository
Purpose (why)	Same	Same	Potentially different
Content (what)	Same	Potentially different	Potentially different
Participants (who/m)	Same	Potentially different	Potentially different
Form (how)	Different	Different	Different
Time (when)	Same	Different	Different
Place (where)	Same	Different	Different
Organizational Consequences	Reinforce and preserve existing communicative practices	Reinforce and enhance existing communicative practices	Potentially transform existing communicative practices

pose, the central identifying characteristic of genre systems, is potentially different—that is, some of these interactions might not have occurred at all before Team Room.

We have noted that the three Mox groups drew on genre systems both habitually and deliberately in coordinating their collaborative work in Team Room. Enacting genre systems (and their constituent genres) habitually is an efficient and easy way for team members to coordinate their actions since it depends on the tacit but common expectations of all team members. Such habitual communicative action allows team members to focus their attention on other aspects of their work. One potential negative consequence of this habitual pattern, however, is that it may mask differing and even conflicting expectations among team members. For example, non-responses to placeholders or circulated drafts (the “what” of the genre system) were differently interpreted (even by the same person) as evidence of both lack of relevant input on the subject and of rudeness or disrespect to the initiator. This downside suggests that groups may initially wish to make assumptions explicit and explore any possible disagreements before moving into a more tacit mode of enactment.

Another, and perhaps more significant, potential downside of drawing on genre systems habitually is the potential for missing opportunities for change. In moving their communication to a new medium, members of a team or community typically import existing genres and genre systems, improvise around them, and gradually learn to take advantage of new

opportunities afforded by the medium (Orlikowski & Yates, 1994). Habitual enactment of old norms in using a new medium reinforces the status quo on all dimensions except for "how." For example, if team members had responded to a circulated draft with individual responses e-mailed to the originator of the draft ("who/m"), they would have reinforced the existing system and hierarchy, failing to take advantage of the medium's capability to allow broader and more interactive participation in collaborative authoring. Although change may occur through slippage in tacitly enacted norms, it occurs more rapidly and readily when individuals deliberately and mindfully draw on and vary structures. For example, the Team Room architecture made it easy for the ISQI team to go beyond the core group to include a large number of peripheral members who might participate in the planning of TQM programs to the extent they wished ("who/m"), a deliberate decision made when they set up the Team Room. That team could have further leveraged this capability to bring new perspectives into the quality initiative by deliberately reaching beyond the IS Department to include, for example, individuals from client departments. Thus habitual enactment of genre systems may forgo opportunities for change along one or more of the dimensions of genre systems.

These potential negative consequences suggest that team members attempt to focus deliberate attention on the various dimensions of genres and genres systems both initially and in an ongoing, if episodic, manner. Constant deliberation on all dimensions of each genre or genre system enacted would be inefficient in coordinating the team's intended activities. But initial attention to these dimensions can help in the transition to a new medium. Further gains, however, are likely only after team members come to have a good working understanding of the new capabilities of the medium. Ideas, such as reaching beyond the IS Department in seeking participation in the ISQI team, are more likely to emerge at a later time than when the medium is initially introduced. We have suggested elsewhere that mediators who are charged with paying deliberate attention to matters of communication in a new medium can play an ongoing and useful role in adapting genres and genre systems to new media (Yates & Orlikowski, 1999).

This paper draws on qualitative data to propose six dimensions around which genre systems carry expectations, and to note that users of a new medium may draw on these expectations either habitually or deliberately. For researchers, we believe that these dimensions provide a helpful way of understanding changes in communicative interaction associated with adoption and use of new electronic media. For users of such new media, we believe that making assumptions explicit that have previously been tacit may facilitate adjustment to the new medium, both to avoid misunderstandings and to encourage experimentation that may lead to change in genre systems.

NOTES

1. We initially and briefly proposed a similar set of dimensions in a paper presented at the Second International Conference on Genre in Vancouver, BC, in 1997 (to be published as Yates & Orlikowski (forthcoming), but we did not focus on, or develop them in that paper.
2. For most of the categories used, intercoder reliabilities measured with Cohen's Kappa ranged from 0.8 to 1, with a few reliabilities between 0.6 and 0.8.
3. Most of the other messages could be categorized into individual genres such as report and announcement.
4. Our language here deliberately invokes what Freedman [1994] terms "uptake," how genres do or do not respond to each other.
5. Lotus Notes was already in use at Mox before Team Room was introduced, so a shift had already begun from e-mail and paper to groupware, with its greater potential for inclusiveness.

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